

Task 1: Adversity Profile Write-up

Overview

This document analyzes the adversity profiles of six clients (A through F) across varying time horizons ($\tau \in \{5, 10, 15, 20, 25, 30\}$ seconds), based on the provided trade execution data. The adversity percentage represents the proportion of trades that are adverse (i.e., result in a negative PnL for the Liquidity Provider) at the closing time $t = \tau$. An adversity rate exceeding 50% indicates that the client possesses a directional edge over the LP.

Observed Trends

1. Universal Positive Correlation with Time Horizon:

Across all six clients, the adversity percentage strictly increases as the time horizon τ extends from 5 seconds to 30 seconds. This trend indicates that adverse selection risk compounds over time. The longer the LP holds the internalized inventory, the higher the probability that the market mid-price drifts against the LP's position.

2. Client Stratification (Informed vs. Uninformed Flow):

The gradient and magnitude of the adversity profiles allow us to stratify the clients into distinct categories based on their toxicity to the LP:

- **Benign / Uninformed Flow (Clients A, B, C):** These clients exhibit low overall adversity that remains well below the 50% threshold at all horizons. For instance, Client A's adversity is relatively flat, rising only from 39.7% to 41.7%. This suggests their order flow is primarily noise or liquidity-driven rather than alpha-driven. Trades with these clients are generally safe to internalize.
- **Marginally Toxic Flow (Client D):** Client D begins with an adversity of 43.3% at $\tau = 5$, but crosses the 50% threshold by $\tau = 25$ seconds (reaching 51.7% at $\tau = 30$). This indicates a medium-term alpha. The LP can likely internalize these trades profitably if they are closed rapidly, but holding them for longer durations leads to adverse selection.
- **Highly Toxic / Informed Flow (Clients E, F):** Clients E and F display highly toxic flow. Client F's adversity starts high (47.0%) and aggressively scales to 61.8% by $\tau = 30$. This steep gradient strongly implies that these clients are "informed traders" possessing a sharp short-term predictive signal. They trade aggressively in the direction of imminent price movements, systematically outperforming the LP's quotes.

Conclusion

The clear divergence in client profiles highlights the necessity of a client-aware dynamic quoting and externalization strategy. Flow from benign clients (A, B, C) can be safely internalized to capture the spread, while trades from toxic clients (E, F) should trigger aggressive externalization mechanisms to mitigate severe adverse selection.